



## Web Development Technologies

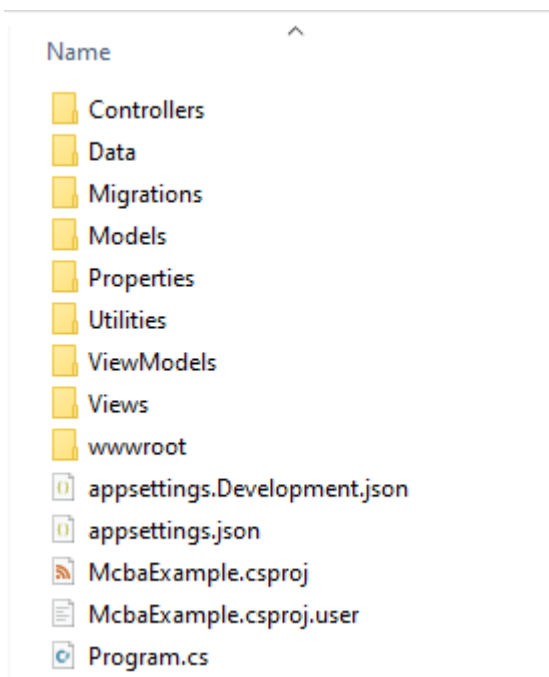
### Tutorial and Lab Sheet 5 Solution Sketch

#### ASP.NET Core MVC Code First - Complex Models

---

### Exercise 1 Creating complex models

Please look at sample solution code archive for this week. Inside **McbaExample** directory you will find these files and folder:



Controller contains HomeController where the first request goes to display account details.

Data folder has the context file.

Models has the 4 model files to create complex models.

ViewModels has files that provide data visualisation to the programmer. It is easier to create ViewModel files than work with Model files at times. The ViewModel is essential when you want a separation of concerns between your DomainModel (Model) and the rest of your code. The less dependencies you have between the Model, View and Controller the easier down the line it will be to make changes to the DomainModel without breaking the interface contracts in the View and Controller.

Spend time looking at HomeController methods especially Index() and second Deposit() methods. In Index method there is a join happening between two entities.



Second Deposit method updates two database tables in one go - Account when money is deposited and Transactions to record a transaction when deposit occurs.

Another interesting thing to look at is- how ViewModel comes into picture to perform some validations such as amount deposited cannot be negative.

Finally look at Views\Home- Index.cshtml and Deposit.cshtml files.